

Highlights

Smart Urban Cooling: Mapping and Mitigating Heat Islands with Physics-Informed Machine Learning

- A novel PINN framework integrates thermodynamics with geospatial data
- Achieves high UHI prediction accuracy ($R^2 = 0.9741$) in Paris case study
- Provides practical tool for green infrastructure cooling assessment
- Introduces reproducible data-fusion protocol for urban climate modelling
- Bridges physical models and data-driven methods for urban planning

Smart Urban Cooling: Mapping and Mitigating Heat Islands with Physics-Informed Machine Learning

Abstract

Mitigating the adverse effects of urban heat islands (UHI) is critical for enhancing climate resilience, public health, and energy sustainability in cities. Traditional modelling approaches, however, often face a trade-off between physical fidelity and the flexibility to integrate diverse geospatial data. This paper presents a novel Physics-Informed Neural Network (PINN) framework that bridges this gap by embedding the heat diffusion equation directly into a neural network, ensuring predictions adhere to thermodynamic principles while learning from complex urban data patterns. The framework includes a systematic data-fusion workflow that unifies multi-format geospatial inputs, such as land surface temperature, vegetation indices, and vector-based urban morphology, into a cohesive gridded structure, enabling scalable and reproducible urban climate analysis. Validated through a case study of Paris, the model achieves exceptional predictive accuracy (R^2 of 0.9741, RMSE = 0.5019°C) and reliably simulates the city's thermal patterns. Beyond diagnosis, it functions as a practical scenario-planning tool, quantifying the cooling effects of green infrastructure interventions such as green roofs. By integrating physical simulation with data-driven analytics, this research provides urban planners and policymakers with a scientifically rigorous, scalable resource to support the design of smarter, cooler, and more sustainable cities.

Keywords: Urban Heat Island, Machine Learning, Physics-Informed Neural Networks, Geospatial Data, Green Infrastructure, Urban Cooling.

1. Introduction

Urban overheating has emerged as a defining challenge for twenty-first century city governance, with profound implications for public welfare, economic productivity, and ecological stability [1, 2, 3]. The growing intensity and frequency of extreme heat events, exacerbated by the urban heat island

(UHI) effect [4, 5, 6], poses critical risks to public health, infrastructure systems, and energy security in cities worldwide. These challenges are further amplified by ongoing urbanisation trends and climate change projections, creating compound risks that demand innovative solutions. Addressing UHI has consequently evolved from a secondary concern to a central pillar of urban resilience and sustainable development strategies [7, 8, 9], particularly as cities strive to meet their climate adaptation commitments under international frameworks.

Green infrastructure (GI), including parks, green roofs, urban forests, and permeable surfaces, is widely recognised as a key nature-based solution for its multifunctional benefits spanning thermal regulation, stormwater management, and biodiversity conservation [10, 11, 12, 13]. The cooling mechanisms of GI operate through multiple pathways, including evapotranspiration processes, shading effects, and altered albedo values, which collectively modify the urban energy balance [14, 15]. Notwithstanding, realising these benefits through effective and cost-efficient deployment requires precise, high-resolution modelling tools capable of both diagnosing current thermal environments and reliably predicting intervention outcomes across diverse urban morphologies and climatic contexts.

Current UHI modelling approaches typically force planners to choose between two problematic paradigms with complementary weaknesses [16]. Data-driven machine learning methods, such as Random Forests [17] or Convolutional Neural Networks (CNNs) [18], demonstrate remarkable capability in identifying complex, non-linear patterns within extensive geospatial datasets. Yet their black-box nature and lack of embedded physical knowledge often yield statistically plausible but thermodynamically inconsistent results, particularly when extrapolating beyond training distributions [19, 20]. This fundamental limitation renders them unreliable for scenarios involving novel urban configurations, changing climate regimes, or unprecedented development patterns, ultimately restricting their value for long-term strategic planning.

Conversely, high-fidelity physical simulations such as numerical models like the Weather Research and Forecasting (WRF) system [21] or computational fluid dynamics (CFD) approaches [22], which are firmly grounded in first principles of fluid dynamics and thermodynamics [23]. While physically robust and scientifically defensible, these models often demand substantial computational resources for rapid city-scale analysis and require specialised expertise that limits their integration into mainstream planning processes. The

parametrisation of urban surfaces and atmospheric processes in these models introduces additional uncertainties, while their substantial computational requirements often preclude the iterative scenario testing needed for evidence-based policy development.

This methodological divide creates a significant implementation gap between academic research and practical application in municipal governance. Although extensive literature demonstrates GI's capacity to mitigate urban heat through observational studies and controlled experiments [24], translating this knowledge into actionable planning guidance remains operationally challenging. Urban planners and policymakers lack practical tools to answer crucial strategic questions with quantitative precision: what magnitude of temperature reductions might different GI typologies achieve at neighbourhood scales under specific climatic conditions, and which urban zones would derive greatest benefit from targeted investments in parks or tree canopies to maximally impact thermal hotspots? This deficiency highlights the urgent need for a hybrid modelling paradigm that merges the adaptive learning capacity of data-driven methods with the rigorous constraints of physical laws while remaining accessible to planning practitioners.

Physics-Informed Neural Networks (PINNs) have emerged as a promising framework to bridge this methodological divide across various scientific domains [25, 26]. By embedding governing partial differential equations directly within neural network loss functions, PINNs can generate solutions that honour both observational data and physical principles, effectively regularising the learning process with domain knowledge. This approach has demonstrated particular value in situations where data may be sparse or noisy, but fundamental physical laws provide strong constraints on possible solutions. Nevertheless, their application to complex urban systems faces substantial practical barriers, particularly regarding the seamless integration of heterogeneous, multi-format geospatial data that characterises urban environments [27, 28].

Urban landscapes comprise diverse data types with varying spatial and temporal resolutions, including raster formats (e.g., satellite-derived land surface temperature and normalised difference vegetation index) and vector data (e.g., building footprints, land use polygons, and transportation networks). Each data type presents unique challenges for standardisation and interoperability within machine learning workflows.

While individual data fusion steps are standard, a reproducible workflow for unifying these disparate data sources within a cohesive PINN framework

for UHI modelling has been underexplored [29], hindering their scalability and adoption in urban science and planning practice [30]. Furthermore, the development of PINN architectures specifically tailored to the multi-scale nature of urban climate processes, spanning from building-scale turbulence to neighbourhood-scale advection, remains an open research challenge.

This paper introduces a novel PINN framework specifically designed for urban climate analysis (Figure 1) that addresses these limitations through three primary contributions:

- A specialised PINN architecture that embeds the heat diffusion equation within a flexible deep learning framework, ensuring thermodynamically consistent UHI predictions while maintaining the pattern recognition capabilities of neural networks.
- A structured data-fusion workflow that systematically translates multi-format geospatial data into unified tensor structures, thereby addressing key interoperability challenges and enabling efficient model training on realistic urban landscapes.
- Comprehensive validation through an extensive Paris case study, demonstrating both predictive accuracy against observational data and practical utility as a decision-support tool for scenario planning and policy evaluation.

In combination, these advancements establish an integrated, scalable approach for high-resolution urban climate modelling that bridges the gap between scientific rigor and practical applicability. The framework provides urban planners and policymakers with a scientifically rigorous yet operationally feasible tool to optimise heat resilience strategies, supporting the development of smarter, healthier, and more sustainable cities in an era of climate uncertainty.

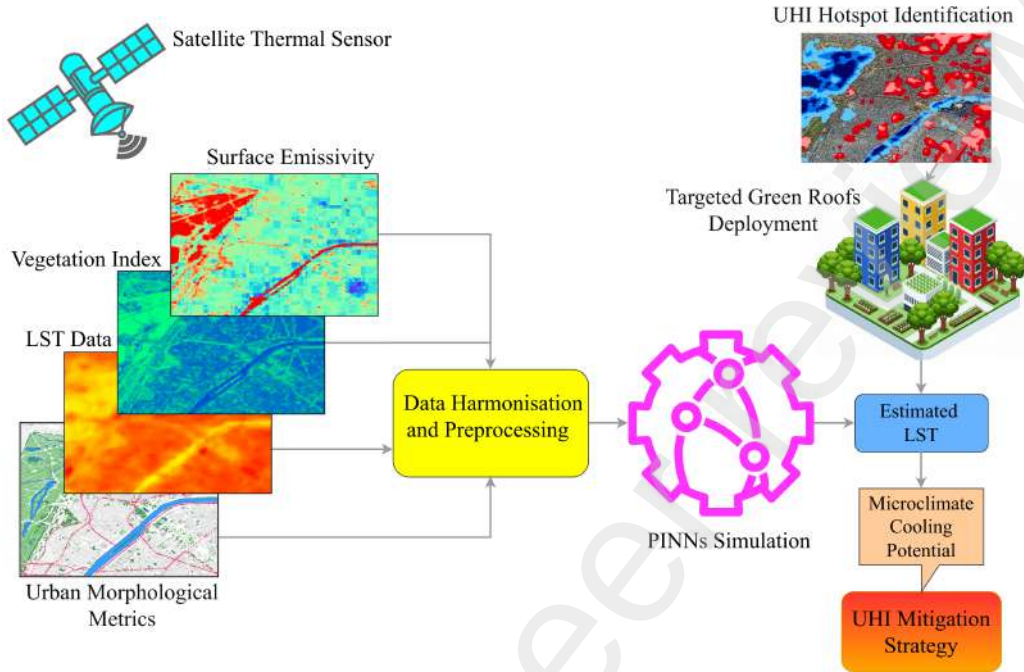


Figure 1: Methodological workflow for UHI analysis and mitigation, integrating physical laws with multi-format geospatial data

2. Dataset and Study Area

2.1. Dataset preparation

The following data fusion protocol transforms heterogeneous geospatial data into a unified input structure suitable for the PINN framework. The process, illustrated in Figure 2, involves three sequential steps designed to preserve data integrity while creating a format compatible with neural network processing.

First, in the gridding step, we reprojected all satellite-derived rasters, specifically Land Surface Temperature (LST), Normalised Difference Vegetation Index (NDVI), and surface emissivity, to a consistent Universal Transverse Mercator (UTM) coordinate system. For instance, data for Paris were reprojected to UTM 32N. We used bilinear resampling to establish a standard, spatially coherent grid at a uniform 30-metre resolution.

The LST was calculated from the thermal infrared band (Band 10) of Landsat 8 Level 2, Collection 2 product: (PRODUCT_ID: LC08_L2SP_199026_20250618_20250627_02_T1). This

derivation followed a two-stage process: initial conversion of top-of-atmosphere radiance (R_{TOA}) to at-sensor brightness temperature (T_{sensor}) using Planck's law with standard Landsat thermal constants, followed by atmospheric correction to retrieve the final LST value [31, 32]. The correction incorporated land surface emissivity (ϵ) and atmospheric water vapour data from NCEP/NCAR reanalysis [33], following the equation:

$$LST = k \left[\frac{1}{\epsilon} (\omega_1 R_{TOA} + \omega_2) + \omega_3 \right] + \eta + T_{sensor} \quad (1)$$

where k and η are calibration parameters derived from the Planck function, while ω_1, ω_2 , and ω_3 represent atmospheric correction coefficients that are functions of total precipitable water vapour [32].

The NDVI was calculated from the Red (B_{Red}) and Near-Infrared (B_{NIR}) bands using the standard formula:

$$NDVI = \frac{B_{NIR} - B_{Red}}{B_{NIR} + B_{Red}} \quad (2)$$

while the theoretical range of the NDVI spans from -1 to 1 , most land surfaces exhibit a much narrower spread of values. To enhance the visual representation of typical conditions within the study area, the index was rescaled to the range $[-0.2, 0.3]$ using Collection 2 parameters [32]. This adjustment improves contrast and interpretability by stretching the critical values: negative readings generally correspond to water, values close to zero indicate bare soil, and positive values reflect increasing vegetation density.

Land surface emissivity was estimated using the NDVI Thresholds Method [32, 34]. Pixels with $NDVI < 0.2$ were classified as bare soil and assigned emissivity values from the ASTER Global Emissivity Dataset (GED). Pixels with $NDVI > 0.5$ were assigned a vegetation emissivity of 0.99 . For mixed pixels ($0.2 \leq NDVI \leq 0.5$), emissivity was calculated as a linear mixture based on the fractional vegetation cover derived from NDVI.

In the rasterisation step, urban morphology vector data from OpenStreetMap [35] were converted from Geographic JavaScript Object Notation (GeoJSON) format [36] into binary raster layers. Each pixel was assigned a value of 1 where a feature was present and 0 where absent, producing rasterised feature maps matching the spatial extent and resolution of the satellite data. The vector-based urban data were classified into distinct categories: impervious surfaces (building footprints and road networks), water bodies, and vegetation types (parks and trees).

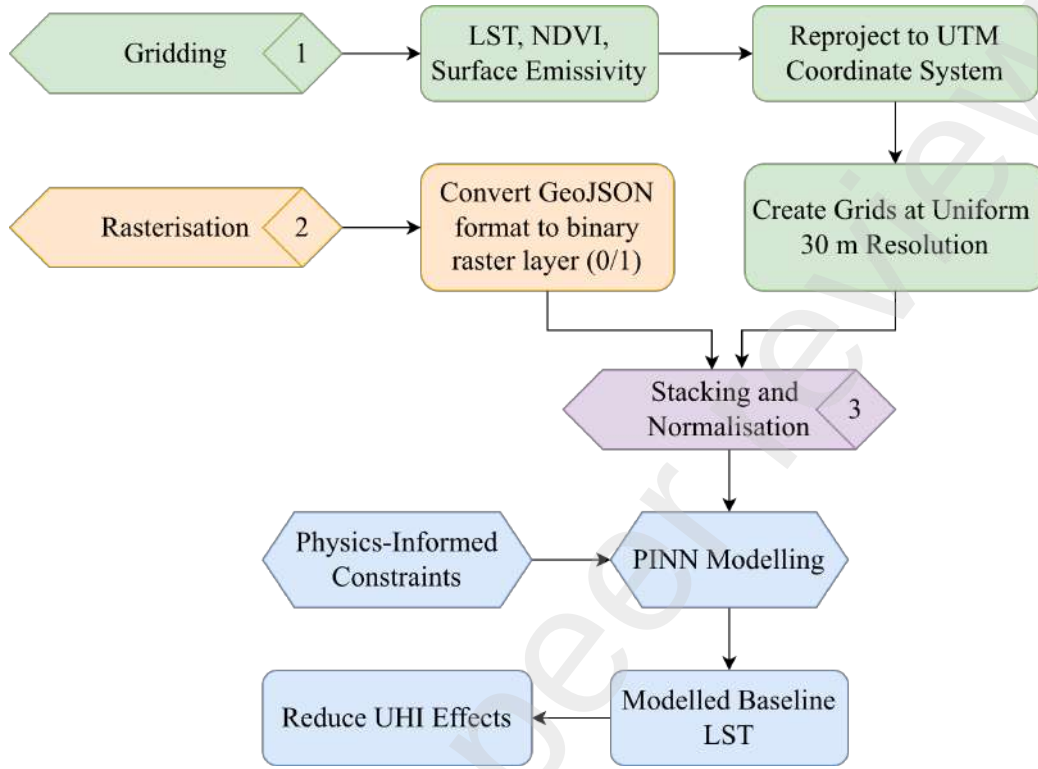


Figure 2: Data fusion and preprocessing workflow for UHI analysis. The protocol integrates heterogeneous geospatial data through three key stages: (1) Gridding, which involves reprojecting raster data to a UTM coordinate system; (2) Rasterisation, which converts vector data into binary layers; and (3) Stacking and Normalisation, which creates a unified input tensor for PINN modelling.

Finally, the stacking and normalisation step combined all aligned layers to create a 10-channel input tensor. This tensor contained spatial coordinates generated from the LST grid, alongside NDVI, emissivity, and six binary urban feature layers. To ensure stable neural network training and maintain numerical consistency, we independently normalised each channel of continuous satellite data to a $[0, 1]$ range using Min-Max scaling. The binary urban feature layers, containing values of either 0 or 1, were used without modification.

2.2. Study area

This research focuses on Paris, France, a densely urbanised European capital facing significant urban heat island challenges [37, 38]. As a city

committed to sustainability and resilience, Paris provides a compelling case study for examining data-driven approaches to smart city planning and climate adaptation .

The study area, shown in Figure 3a, covers central Seine-Paris with coordinates spanning 2.26–2.31°E and 48.85–48.88°N. This bounding box was selected to capture the gradient from the dense urban core to peripheral areas with more vegetation, thereby encompassing the full spectrum of urban thermal environments. The analysis period of 15–30 June 2025 captures characteristic summer conditions when urban heat intensity peaks, creating particular challenges for public health and energy management.

Panel (b) presents the classified urban morphology, distinguishing six key features: parks, water surfaces including the Seine River, tree canopies, road networks, rail corridors, and built structures. This classification reveals the complex interplay between natural and built environments that drives thermal and ecological patterns across the city.

The LST map in panel (c) shows a striking UHI signature, with values ranging from 27.7°C (yellow) in cooler areas to 44.2°C (red) in thermal hotspots. The warmest zones concentrate in densely built areas, while cooler regions correspond to water bodies and vegetated parks, demonstrating how urban form directly influences local climate conditions.

Vegetation distribution, quantified through NDVI in panel (d), ranges from -0.52 (dark blue) in water and built areas to 0.91 (green) in densely vegetated zones. The spatial pattern reveals an inverse relationship with LST, where higher NDVI values in parks and tree canopy areas correspond with lower temperatures, highlighting vegetation's cooling role in the urban environment.

Surface emissivity variations in panel (e) range from 0.96 (blue) to 0.99 (red), reflecting differences in surface materials. Built structures such as concrete typically exhibit lower emissivity, while natural surfaces like vegetation and water show higher values. These variations are crucial for accurate thermal radiation modelling and LST retrieval from remote sensing data.

The kernel density estimates in panels (f)–(h) provide statistical characterisation of these variables for model training. The bimodal LST distribution in panel (f) shows a dominant peak at 36–40°C representing urban built-up areas, and a secondary peak at 28–32°C corresponding to cooler natural features. The NDVI distribution in panel (g) features a primary peak at 0.15–0.25 indicating moderate vegetation coverage, and a secondary peak at 0.7–0.8 representing dense parkland vegetation. Panel (h) shows emissivity

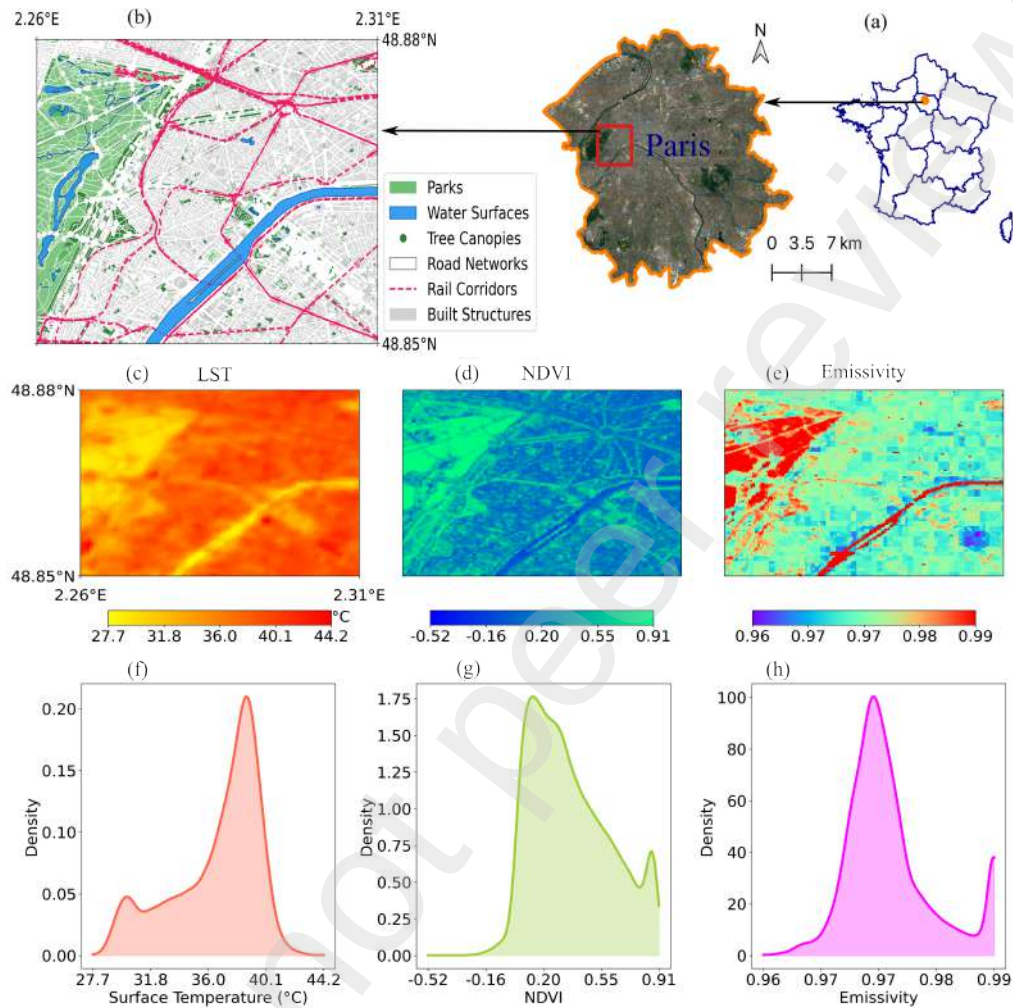


Figure 3: (a) Geographic context of Paris with study area indicated (red box) and 7 km scale bar; (b) Urban morphology classification showing parks, water bodies, tree canopies, transportation networks and built structures; (c–e) Spatial distributions of LST, NDVI and surface emissivity; (f–h) Kernel density estimates showing statistical distributions of each variable for model training.

concentrated around 0.97 for typical urban materials, with a minor peak near 0.99 for natural surfaces. Taken together, these panels document the spatial patterns and statistical distributions essential for robust model training. The integrated analysis bridges geographic, remote sensing, and statistical approaches, providing a comprehensive foundation for urban climate research

and heat mitigation strategies in smart city contexts.

3. Methodology

The high-resolution modelling of the UHI effect is achieved through a PINN. This framework represents a core innovation by seamlessly integrating thermodynamic principles with remote sensing data within a single, differentiable learning system [39, 40]. Its development is built upon three pillars: the network architecture, the governing physical formulation, and the training protocol.

3.1. PINN Architecture and Physical Integration

The proposed PINN model learns a mapping from a suite of geospatial features to the LST field, constrained by the laws of physics. As Figure 4 illustrates, the network is a fully connected deep neural network that accepts a 10-dimensional input vector for each pixel and outputs a single scalar value: the predicted LST, denoted as $T^*(x, y)$.

3.1.1. Input Feature Engineering

The input vector is carefully curated to encapsulate the primary factors influencing the surface energy balance:

- Spatial coordinates (x, y) : These allow the model to capture inherent spatial patterns and autocorrelations in the UHI phenomenon.
- NDVI: This serves as a proxy for vegetation density and its associated evapotranspirative cooling effect.
- Surface emissivity: A critical parameter for accurate thermal radiation modelling, derived from land cover classifications.
- Urban morphology features: Six features quantifying the density of key urban elements, normalised to a $[0,1]$ scale: building footprints, road network, railway infrastructure, surface water, park areas, and tree canopy cover. These inform the model about surface properties that directly affect heat storage and release.

3.1.2. Network Design

The network comprises 4 hidden blocks, with the number of neurons progressively decreasing from 256 to 32 (detailed in Table 3). This funnel-like architecture is designed to first capture complex, high-dimensional interactions in the input data before refining them into a physically consistent output.

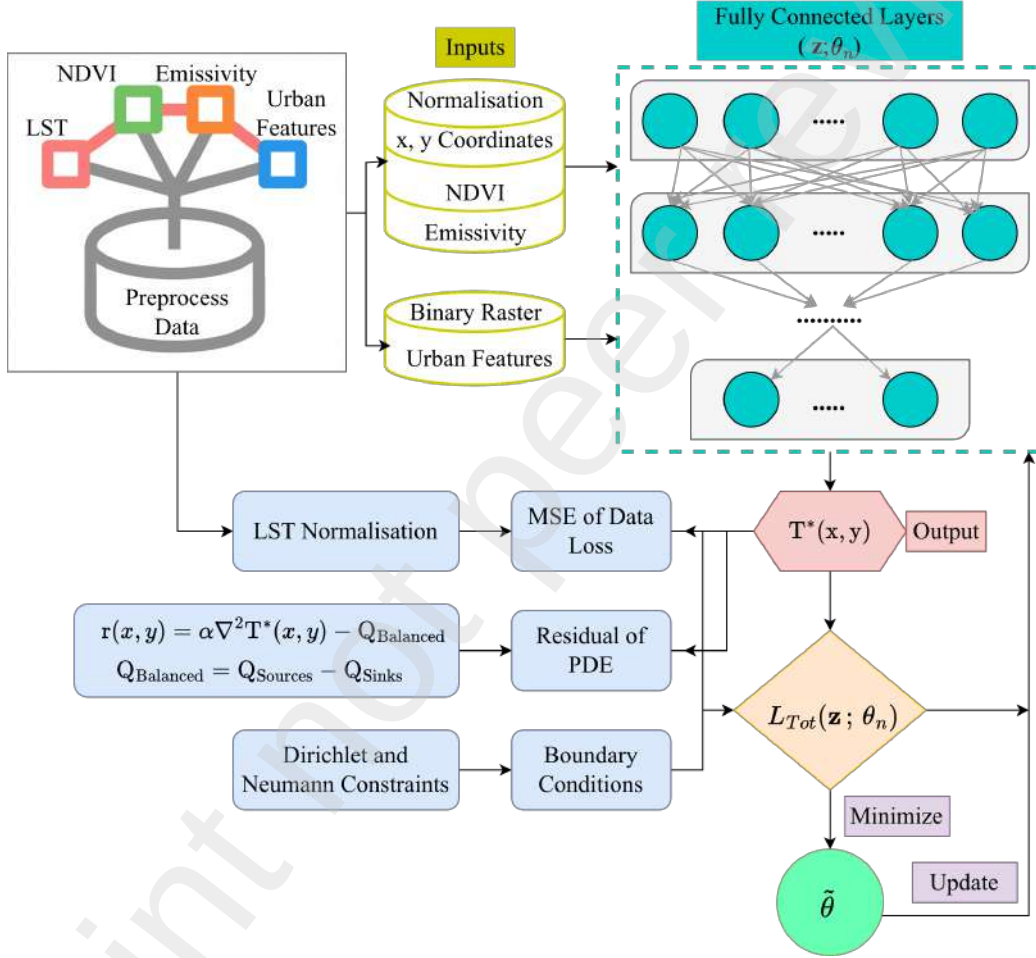


Figure 4: Architecture of the PINN for UHI modelling, featuring a 256→128→64→32 neuron structure (Table 3). The network processes 10 geospatial features per pixel through a composite loss function (L_{Total}) that balances satellite data fidelity with physical consistency via the heat equation and boundary conditions. The model parameters θ are optimised to ensure the predicted LST field T^* is both data-accurate and thermodynamically consistent.

Activation functions were selected for their specific properties:

- The hyperbolic tangent (Tanh) function, used in the initial and physics-proximal layers, provides smooth, bounded derivatives. This makes it ideal for modelling continuous physical fields like temperature and ensures stable computation of higher-order derivatives via automatic differentiation.
- The LeakyReLU function (negative slope=0.01), employed in intermediate layers, helps to mitigate the vanishing gradient problem. It promotes effective learning of complex, non-linear feature interactions without the complete deactivation of neurons that can occur with the standard ReLU function.

The hybrid activation function combination addresses the vanishing gradient problem in deep networks whilst maintaining the smoothness required for physics-based loss terms [41, 42]. This approach enables effective information flow during backpropagation while preserving accurate representation of continuous physical phenomena. For comparison, Table 2 also includes results using a single Tanh activation function.

3.2. Loss Formulation

The novelty of the PINN approach is encapsulated in its composite loss function, which regularises the learning process with physical laws [40]. The total loss, L_{Total} , is a weighted sum of three critical components:

$$L_{\text{Total}} = \gamma_0 \cdot L_{\text{Data}} + \gamma_1 \cdot L_{\text{PDE}} + \gamma_2 \cdot L_{\text{BCs}} \quad (3)$$

3.2.1. Data Loss (L_{Data})

This component ensures the model's fidelity to empirical measurements. It is defined as the Mean Squared Error (MSE) between the model's predictions and the satellite-derived LST at all interior data points:

$$L_{\text{Data}} = \frac{1}{N_d} \sum_{j=1}^{N_d} [T^*(x_j, y_j) - T_{\text{obs}}(x_j, y_j)]^2 \quad (4)$$

where N_d is the number of interior pixels with available LST data.

3.2.2. Physics Loss (L_{PDE})

The physics-informed loss incorporates regularisation through a simplified heat equation formulation:

$$\alpha \nabla^2 T^* = Q_{\text{Balanced}} \quad (5)$$

where $\nabla^2 T^*$ represents the Laplacian of the temperature field (computed via automatic differentiation), α denotes the spatially varying thermal diffusivity, and Q_{Balanced} captures the net heat exchange between urban heat sources and sinks.

This formulation conceptualises the urban environment as a bulk layer with effective material properties, following a well-established approach in urban climate modelling [43]. Similar simplifications underpin models like UrbClim, which represents urban surfaces as rough impermeable slabs. This strategy strikes a practical balance between physical realism, computational efficiency, and applicability in data-scarce contexts [43]. Evidence further supports that such urban parameterisations significantly enhance surface temperature representation compared to non-urbanised models [44].

At this study's 30m resolution, spatial variations in thermal diffusivity α remain relatively modest compared to temperature gradients [44, 45]. We therefore omit the advection term $\nabla \alpha \cdot \nabla T^*$, focusing instead on the dominant diffusion processes that characterise urban heat patterns at this scale [41].

- **Thermal Diffusivity (α):** We define α as a pixel-wise parameter based on land cover composition, reflecting the material properties of the urban fabric. Values are assigned from established literature on urban material properties [46]: high for impervious surfaces ($\alpha_{\text{building}} \sim 1.0 \times 10^{-6} \text{ m}^2/\text{s}$), moderate for vegetation ($\alpha_{\text{veg}} \sim 0.33 \times 10^{-6} \text{ m}^2/\text{s}$), and low for water bodies ($\alpha_{\text{water}} \sim 0.19 \times 10^{-6} \text{ m}^2/\text{s}$). This classification aligns with studies showing strong thermal diffusion effects from urban heat sources and sinks [47]. The higher thermal diffusivity of dense urban materials like concrete and asphalt compared to vegetation is well-documented [48], consistent with established urban parameterisations where impervious surfaces facilitate faster heat conduction despite their greater heat retention capacity. For mixed pixels, we compute a weighted average based on the fractional coverage of each land cover type.
- **Net Heat Exchange (Q_{Balanced}):** This balanced heat exchange term captures the equilibrium between anthropogenic heat sources and natural

cooling sinks through the formulation:

$$Q_{\text{Balanced}} = Q_{\text{Sources}} - Q_{\text{Sinks}} \quad (6)$$

This formulation offers a physically realistic view of the urban energy balance by including both heating and cooling processes. It is designed to be interpretable, using distinct methods that prevent circularity with the model's other binary features. The approach also allows the resulting energy balance patterns across the study area to be visually verified.

The heat sources (Q_{Sources}) combine several anthropogenic contributions, weighted according to established urban climate literature [49, 50]:

$$Q_{\text{Sources}} = 0.30 \cdot \text{BD}_{\text{norm}} + 0.25 \cdot \text{RD}_{\text{norm}} + 0.20 \cdot \text{ID}_{\text{norm}} + 0.15 \cdot \text{UC}_{\text{norm}} \quad (7)$$

Equation 7 incorporates four key urban heat sources. Building density (BD_{norm}) captures heat from buildings themselves, including HVAC systems and reduced ventilation between structures. We calculate this using kernel density estimation smoothed with a 60-metre radius ($\sigma = 2.0$), which reflects the typical downwind influence zone of two to three building heights [4]. Road density (RD_{norm}) accounts for traffic emissions and asphalt heat storage, derived from road network data smoothed with a 45-metre radius ($\sigma = 1.5$) to represent localised heat plumes [51]. Proximity to major roads (ID_{norm}) uses an inverse distance transform from motorways, trunks, and primary roads. Finally, urban canyon effects (UC_{norm}) capture trapped longwave radiation and reduced sky visibility in dense urban areas, quantified using building perimeter-to-area ratios smoothed with a 30-metre radius ($\sigma = 1.0$) to represent these hyper-local phenomena.

The heat sinks (Q_{Sinks}) capture natural cooling mechanisms and are composed as follows:

$$Q_{\text{Sinks}} = 0.15 \cdot \text{PD}_{\text{norm}} + 0.10 \cdot \text{WD}_{\text{norm}} + 0.10 \cdot \text{TD}_{\text{norm}} + 0.10 \cdot \text{CP}_{\text{norm}} \quad (8)$$

The four components in Equation 8 work together to mitigate urban heat through different natural processes. Park density (PD_{norm}) quantifies cooling from evapotranspiration, shading, and improved air circulation in green spaces, computed with a 90-metre radius ($\sigma = 3.0$) to account

for their extended cooling influence [24]. Water body density (WD_{norm}) captures evaporative cooling and high thermal inertia from rivers and lakes, using a 60-metre radius ($\sigma = 2.0$). Tree canopy density (TD_{norm}) represents shading and transpiration from urban trees, also calculated with a 60-metre radius ($\sigma = 2.0$). Finally, cooling proximity (CP_{norm}) models advective cooling from nearby green and blue spaces, calculated using an inverse distance transform: $CP_{\text{norm}} = (1 + d_{\text{cooling}})^{-1}$.

A Gaussian filter with specific σ values is applied to each feature (in Eqs. 7 & 8) to define its spatial influence, guided by urban climate studies [4, 51, 24]. Generally, heat sources like buildings and roads use smaller sigmas (1.5–2.5) to reflect their localised impact. Alternatively, heat sinks like parks use larger sigmas (2.5–3.5) to represent their broader cooling effect. Very localised features, such as urban canyons, use the smallest sigmas (0.5–1.5). All components then are normalised before their weighted combination to ensure a physically realistic representation of the urban energy balance, helping the model distinguish net heat sources in urban cores from net sinks in parks and water bodies.

Figure 5 presents a comprehensive spatial analysis of urban heat dynamics through three integrated plots. Each plot visualises normalised raster layers to quantify heat sources, heat sinks, and their net balance. Row 1 (a–d) details the heat sources, where warmer colours (yellow to red) indicate higher contributions: (a) building density reveals intense heat generation in densely built zones, for example the bright yellow core in the northwest, reflecting anthropogenic heat emissions and thermal storage from structures; (b) road density highlights linear corridors of high heat (red), corresponding to major thoroughfares where traffic and asphalt storage dominate; (c) inverse distance to major roads intensifies near road networks, such as the central red band, emphasising proximity-driven heat accumulation from vehicle emissions; and (d) urban canyon shows fragmented high-heat zones (yellow patches) where narrow streets and tall buildings reduce sky view, trapping heat through reduced ventilation. These sources together illustrate how built infrastructure amplifies urban warming.

In contrast, row 2 (e–h) illustrates the heat sinks, using cooler colours (blue to purple) for higher cooling capacity: (e) park density identifies large, contiguous cooling zones, such as the purple clusters in the southwest, leveraging evapotranspiration and shading; (f) water density

pinpoints linear cooling corridors, for example the blue river in the center, where high heat capacity and evaporative cooling mitigate temperatures; (g) tree density displays a mixed pattern, with dense tree cover (purple) in parks and fragmented urban areas enhancing transpiration; and (h) cooling proximity integrates all sink features, showing broad purple zones where access to parks, water, or trees amplifies cooling effects. Notably, sink distribution is more spatially heterogeneous than sources, with parks and water surfaces acting as critical "cool islands" amid urban development.

Synthesising the contributions from rows 1 and 2, row 3 quantifies the resulting urban thermal balance. The combined heat sources (i) layer spans 0.14 (pale yellow) to 0.78 (deep red), with the highest intensities (0.62–0.78) concentrated in the northwestern and central urban core, reflecting areas of dense infrastructure where building density, road networks, and urban canyon effects maximise heat generation. The combined heat sinks (j) layer ranges from 0.00 (cyan) to 0.34 (magenta), where the maximum value (0.25–0.34) appears in the southwestern quadrant, corresponding to zones with high park density, water bodies, and tree cover that enable evaporative cooling and shading; the minimum value (0.00) indicates areas with negligible cooling capacity, such as fragmented or built-up zones.

The balanced Q (in k panel), representing the net heat balance (sources minus sinks), exhibits the broadest range: -0.16 (dark blue) to 0.72 (red). Negative values, for example -0.16 in the southwest, confirm localised cooling dominance where sinks exceed sources, while extreme positive values (0.50–0.72) in the central-northern districts indicate severe heat surplus. The full spectrum of this Q term reveals a city where thermal conditions vary from net cooling (blue zones) to intense residual heating (red zones), with the numerical range explicitly demonstrating the spatial heterogeneity of urban heat vulnerability and resilience. Overall, Figure 5 demonstrates that while built environments drive localised heating, strategic integration of sinks can counterbalance these effects, offering a data-driven framework for urban climate resilience planning.

Appendix A presents a sensitivity analysis using alternative weight configurations. These include scenarios for enhanced cooling (through increased green infrastructure), extreme urban heat (maximum UHI), as well as transport-dominated, building-dominated, and water-focused

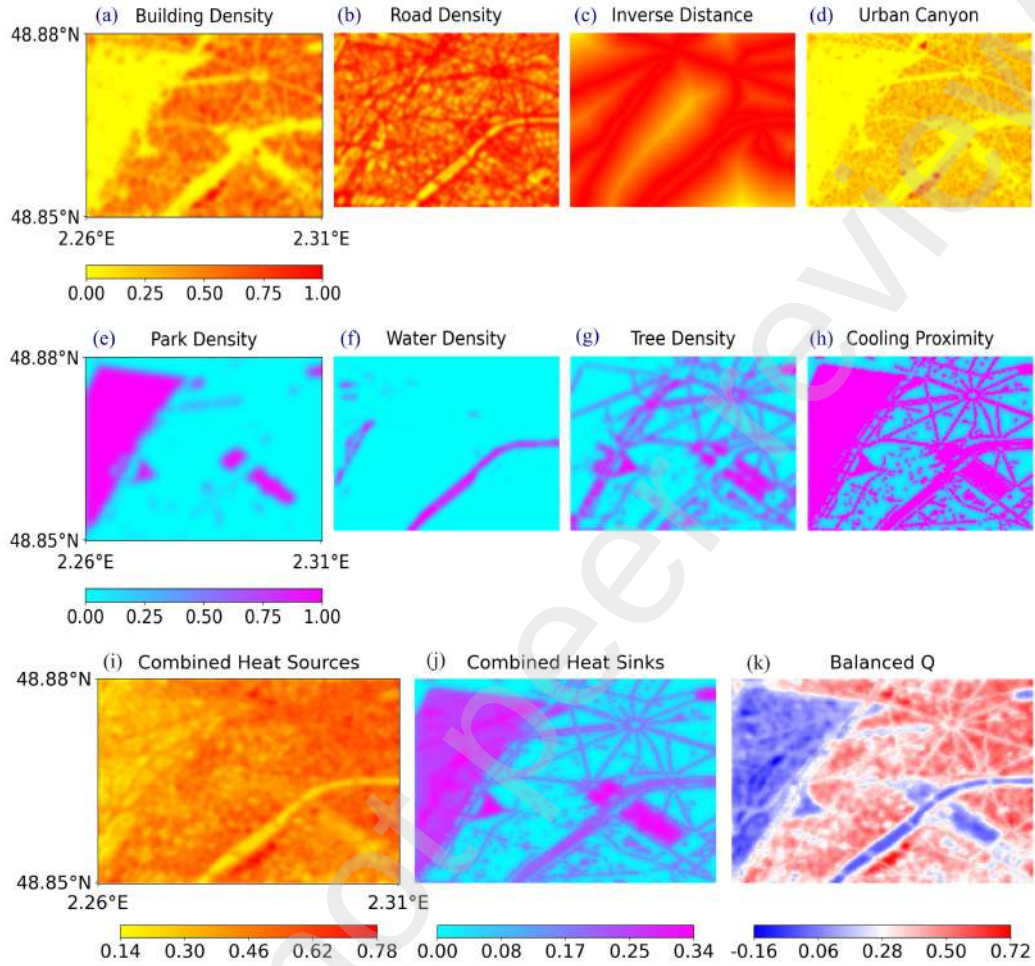


Figure 5: (a–d): Normalised urban heat source components: building density, road density, proximity to major roads, and urban canyon effect; (e–h): Normalised urban heat sink components: park density, water body density, tree density, and proximity to cooling features; (i–k): Combined urban heat balance: weighted heat sources minus heat sinks produces the final balanced Q field.

cooling. The results demonstrate the model's robustness across a range of UHI intensities.

The Physics Loss is then the MSE of the PDE residual $r(x,y) = \alpha \nabla^2 T^* - Q_{\text{Balanced}}$, evaluated at a set of N_p collocation points sampled across the

domain:

$$L_{\text{PDE}} = \frac{1}{N_p} \sum_{j=1}^{N_p} [r(x_j, y_j)]^2 \quad (9)$$

3.2.3. Boundary Condition Loss (L_{BC})

To ensure a well-posed solution, physically realistic constraints are applied at the domain boundaries, defined as the outermost 5-pixel perimeter.

- Dirichlet constraints: Fixed temperature values, derived from satellite observations, are enforced along urban boundaries to anchor the model to known values [52].
- Neumann constraints: Zero-heat-flux requirements ($\partial T / \partial n = 0$) are imposed along boundaries adjacent to large natural features (e.g., the Seine River, major parks) where the domain transitions to areas with significantly different thermal characteristics. This is a common and justified simplification in urban-scale thermal modelling [53] for a domain of this scale, where the primary focus is on resolving intra-urban temperature gradients rather than the absolute boundary heat fluxes.

The loss then is computed as the MSE of the deviations from these specified boundary values and gradients.

3.3. Model Configuration and Training

The final model architecture, summarised in Table 1, reflects design choices informed by computational efficiency and physical modelling requirements. The progressive reduction in layer dimensions (256, 128, 64, 32 neurons) follows an encoder-decoder philosophy, gradually compressing high-dimensional feature representations into the scalar temperature field while preserving physical constraints. This architectural strategy balances model capacity with generalisation, preventing overfitting while maintaining sufficient expressiveness for complex urban thermal patterns.

The model was implemented using the PyTorch framework [54], selected for its robust automatic differentiation capabilities essential for computing PDE residuals in the physics loss term. The Adam optimiser [55] provided adaptive learning rates that proved particularly beneficial for navigating the complex loss landscape of the composite objective function. Training proceeded for 200 epochs with a batch size of 512, configurations determined

Table 1: Detailed PINN architecture and training configuration for urban thermal modelling.

Category	Component / Parameter	Configuration / Details
Architecture	Input Layer	10 neurons (x, y, NDVI, Emissivity, 6 urban features)
	Hidden Blocks	Input Block: 2 layers (256, 256), Tanh Middle Block: 2 layers (256, 128), LeakyReLU (slope=0.01) Physics Block: 2 layers (128, 64), Tanh Output Block: (32, Tanh) $\rightarrow$ (1, Linear)
Training	Optimiser	Adam
	Learning Rate	0.001
	Batch Size	512
	Epochs	200
	Loss Weights	γ_0 (Data) = 1.0, γ_1 (PDE) = 0.1, γ_2 (BCs) = 0.3
Summary	Trainable Parameters	194,177

through empirical testing to provide stable convergence without excessive computational overhead.

The training protocol maintained the optimised loss weights identified during hyperparameter tuning ($\gamma_1 = 0.1$ for physics, $\gamma_2 = 0.3$ for boundary conditions), with the data loss ($\gamma_0 = 1.0$) serving as the baseline reference. This configuration, comprising 194,177 trainable parameters, successfully learned to map urban morphological features to temperature distributions while respecting underlying heat transfer physics.

Selecting appropriate loss weighting coefficients and activation functions presented a critical challenge in balancing the PINN’s data-driven accuracy with physical consistency. A systematic evaluation determined the optimal trade-off between these competing objectives, focusing specifically on the physics loss weight (γ_1) and boundary condition loss weight (γ_2). A sensitivity analysis of these hyperparameters is provided in Subsection 3.5.

3.4. Model Validation Metrics

To comprehensively assess model performance beyond the training regime, the PINN’s predictive accuracy was evaluated on a completely held-out test set

using three complementary statistical metrics. This multi-faceted evaluation approach captures different aspects of prediction quality:

The coefficient of determination (R^2) quantifies the proportion of variance in observed temperatures explained by the model, providing insight into its overall predictive power:

$$R^2 = 1 - \frac{\sum_{j=1}^M [T_{\text{obs}}(j) - T^*(j)]^2}{\sum_{j=1}^M [T_{\text{obs}}(j) - \mu_{\text{obs}}]^2} \quad (10)$$

where μ_{obs} represents the mean observed LST, $T_{\text{obs}}(j)$ denotes the observed LST for pixel j , and $T^*(j)$ indicates the predicted LST.

The root mean square error (RMSE) emphasises larger prediction errors through squaring, making it particularly sensitive to outliers and extreme values:

$$\text{RMSE} = \sqrt{\frac{1}{M} \sum_{j=1}^M [T_{\text{obs}}(j) - T^*(j)]^2} \quad (11)$$

The mean absolute error (MAE) offers a more robust measure of typical prediction error magnitude, being less influenced by occasional large errors:

$$\text{MAE} = \frac{1}{M} \sum_{j=1}^M |T_{\text{obs}}(j) - T^*(j)| \quad (12)$$

Across all metrics, $M = 27,738$ represents the total number of pixels in the independent test set. This comprehensive validation approach ensures that model performance is assessed from multiple perspectives, with higher R^2 values, and lower RMSE and MAE scores cumulatively indicating superior predictive capability and model reliability [56].

3.5. Optimising Sensitivity Hyperparameters

Achieving optimal balance between the PINN's data-driven accuracy and physical consistency required careful tuning of loss weighting coefficients. Our analysis evaluated weights for the physics loss (γ_1) and boundary condition loss (γ_2) using a hybrid Tanh-LeakyReLU activation function, alongside a Tanh-only baseline for comparison, while maintaining other hyperparameters as specified in Table 1.

Results in Figure 6 and Table 2 reveal the model's sensitivity to γ_1 and γ_2 values. Test 1 ($\gamma_1 = 0.1$, $\gamma_2 = 0.3$) showed rapid loss reduction over 10 epochs

but premature convergence, yielding suboptimal performance ($R^2=0.8541$, RMSE=1.1913°C).

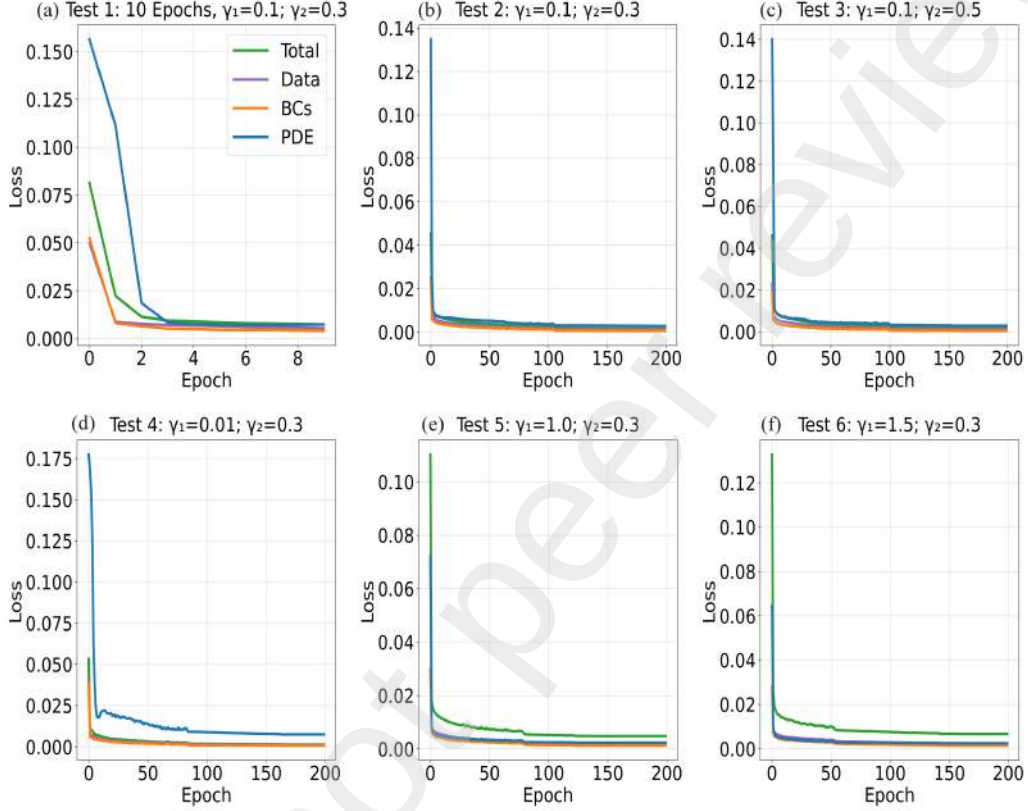


Figure 6: Influence of physics (γ_1) and boundary condition (γ_2) loss weights on training convergence. Configuration (b) achieves optimal balance, while insufficient training (a) or excessive physics weighting (e-f) degrade performance.

In contrast, Test 2 (identical weights but trained for 200 epochs) demonstrated stable, synchronised convergence across all loss components (Figure 6b), achieving superior metrics ($R^2=0.9741$, RMSE=0.5019°C). This confirms that sufficient training time allows this weighting to effectively balance the model's objectives.

Deviating from these optimal weights consistently reduced performance. Increasing γ_2 to 0.5 (Test 3) lowered R^2 to 0.9697, while reducing γ_1 to 0.01 (Test 4) slowed physics-loss convergence and marginally decreased accuracy ($R^2=0.9701$).

Overemphasising physics loss with γ_1 values of 1.0 or 1.5 (Tests 5 and 6) disrupted training balance, producing unstable, oscillatory loss trajectories (Figure 6e-f) and significantly worse performance. Test 6 achieved only $R^2=0.9311$ with $\text{RMSE}=0.8188^\circ\text{C}$ as high physics weighting overshadowed data-driven learning.

The Tanh-only model (Table 2) showed substantially worse metrics than Test 2 ($R^2=0.8357$ versus 0.9741), confirming that activation function choice independently impacts performance even with optimal loss weights.

Test 2's configuration ($\gamma_1 = 0.1$, $\gamma_2 = 0.3$) thus represents the most effective trade-off, producing a physically consistent and highly accurate model. These findings demonstrate how subtle weight adjustments directly influence both training stability and predictive performance.

Table 2: Optimized model configurations and performance. Observed LST range: $27.7\text{--}44.2^\circ\text{C}$.

Test	γ_1	γ_2	R^2	RMSE ($^\circ\text{C}$)	MAE ($^\circ\text{C}$)	Predicted Range ($^\circ\text{C}$)	Max Error ($^\circ\text{C}$)
1	0.1	0.3	0.8541	1.1913	0.8846	28.9–40.6	7.09
2	0.1	0.3	0.9741	0.5019	0.3723	27.7–43.9	4.53
3	0.1	0.5	0.9697	0.5426	0.3992	28.0–43.9	4.45
4	0.01	0.3	0.9701	0.5390	0.3964	27.8–45.9	6.79
5	1.0	0.3	0.9428	0.7462	0.5501	28.6–42.3	6.35
6	1.5	0.3	0.9311	0.8188	0.5992	27.9–41.9	7.01
Only Tanh activation	0.1	0.3	0.8357	1.2642	0.9433	29.0–41.5	7.49

4. Results and Discussion

4.1. Spatial Prediction Accuracy of the PINN Model

The PINN model demonstrated exceptional predictive capability for the summer conditions in the Paris case study, achieving an R^2 of 0.9741 alongside an RMSE of 0.5019°C and MAE of 0.3723°C . It is noteworthy that a related architecture achieved similarly high performance ($R^2 > 0.92$) in a prior study of a different city [34]. This consistent performance across distinct urban environments reinforces the robustness and transferability of the core methodology.

Figure 7 presents a comprehensive visual assessment of the optimal model configuration, revealing strong spatial correspondence between predicted and

observed thermal patterns. The model successfully reproduces the characteristic urban heat island structure in panels (a) and (b), with temperature ranges closely matching observational data (27.7–43.9°C predicted versus 27.7–44.2°C observed). The spatial error distribution in panel (c) shows minimal deviations across most of the domain with slightly elevated errors concentrated in areas of complex urban morphology where microclimatic interactions present greater modelling challenges.

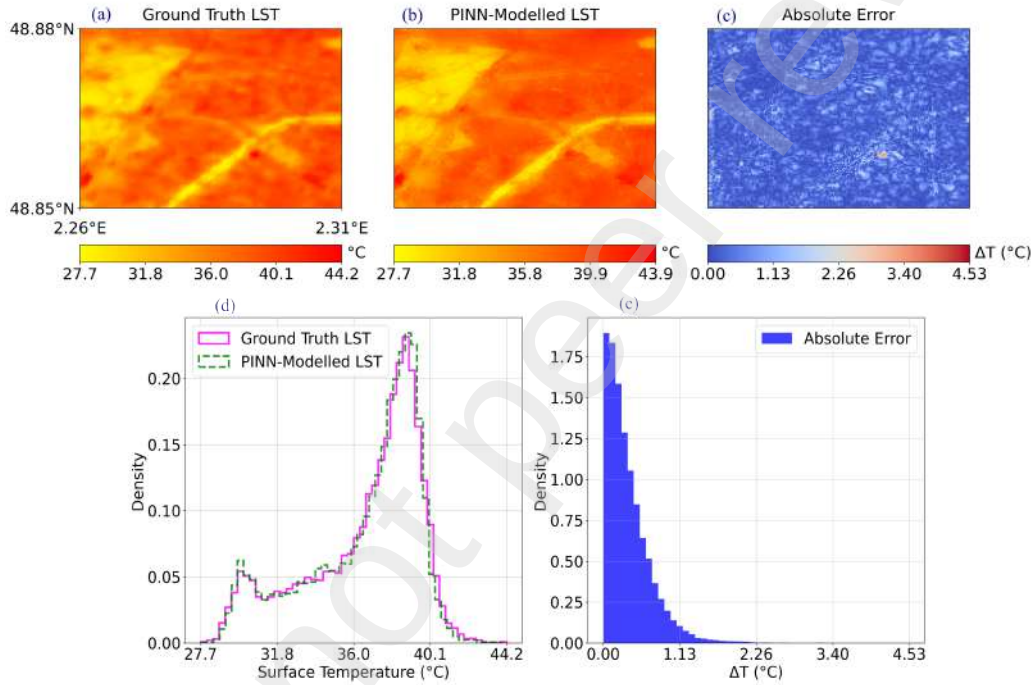


Figure 7: Comprehensive validation of the optimal PINN model ($\gamma_1 = 0.1$, $\gamma_2 = 0.3$, 200 epochs). (a) Satellite-derived ground truth LST. (b) PINN-predicted LST distribution. (c) Spatial pattern of absolute prediction error. (d) Density scatter plot comparing observed versus predicted values. (e) Histogram distribution of absolute error.

Notably, these elevated errors predominantly occur in areas characterised by very high building density, complex urban canyon geometry, and mixed land use patterns. Specifically, the highest prediction errors (2.26–4.53°C) cluster in the central business district and dense residential quarters where the 30-metre resolution cannot fully resolve the intricate shadow patterns, wind channelling effects, and heterogeneous surface materials that define street-level microclimates [53]. The model’s 2D formulation also struggles to capture the

full complexity of 3D radiative trapping in deep urban canyons, leading to systematic underestimation of peak temperatures in these morphologically complex zones.

Statistical validation through density analysis further confirms the model's robustness. The tight clustering along the unity line in panel (d) demonstrates consistent predictive accuracy across the entire temperature range, while the error distribution in panel (e) shows a strong peak near zero, aligning with the low MAE. This performance reflects the balanced training dynamics observed during hyperparameter optimisation, where the composite loss achieved stable convergence by effectively reconciling data fidelity with physical constraints. The model's success in capturing both the spatial structure and statistical distribution of urban temperatures underscores its value as an analytical tool for urban heat island studies. By maintaining physical consistency while achieving high predictive accuracy, the PINN framework provides a reliable approach for mapping thermal environments across complex urban landscapes.

4.2. Comparative Model Performance

Table 3 compares the performance of different model configurations for temperature prediction against the true observed range of 27.7–44.2°C.

Table 3: Model performance comparison

Test Configuration	R^2	RMSE (°C)	MAE (°C)	Range (°C) Predicted	Max Error (°C)
<i>True temperature range: 27.7-44.2°C</i>					
Only data loss	0.9717	0.5250	0.3479	27.7-45.1	6.65
Only data and PDE loss	0.9608	0.6174	0.4321	27.8-44.9	7.79
Data, PDE and BCs loss	0.9741	0.5019	0.3723	27.7-43.9	4.53
Random Forest (200 trees)	0.9673	0.5642	0.3662	28.5-42.5	6.67
Random Forest (500 trees)	0.9676	0.5612	0.3634	28.5-42.6	6.62
Random Forest (1000 trees)	0.9677	0.5602	0.3623	28.5-42.5	6.69

Among the PINN configurations, the model incorporating "Data, PDE and BCs loss" delivered the strongest performance. It achieved the highest R^2 (0.9741), lowest RMSE (0.5019°C), and smallest maximum error (4.53°C). Its predicted temperature range (27.7–43.9°C) also aligned most closely with observations, demonstrating robust adherence to physical constraints. This configuration, which weighted boundary condition loss ($\gamma_2 = 0.3$) more heavily

than PDE loss ($\gamma_1 = 0.1$) alongside data loss ($\gamma_0 = 1.0$), outperformed both the "Only data loss" model ($R^2 = 0.9717$, max error = 6.65°C) and the "Only data and PDE loss" variant ($R^2 = 0.9608$, max error = 7.79°C). The performance degradation in the latter case, where PDE loss was added without boundary conditions, underscores the critical stabilising role of boundary constraints.

To benchmark the PINN against a established data-driven method, we compared it with Random Forest (RF) models. As a widely used algorithm for spatial prediction tasks [57, 58], RF provides a strong baseline for a purely data-driven, non-physics-based approach [59]. Nevertheless, all RF models (with 200, 500, and 1000 trees) exhibited lower R^2 values (0.9673–0.9677), higher RMSE ($\sim 0.56^\circ\text{C}$), and a consistently narrowed predicted range ($28.5\text{--}42.5^\circ\text{C}$) that failed to capture the true temperature extremes. Their maximum errors ($6.62\text{--}6.69^\circ\text{C}$) were also substantially higher than the best PINN configuration. Notably, increasing the number of trees yielded negligible improvement, highlighting a core limitation of the purely data-driven approach for this task. These results collectively show that integrating physics-based constraints, particularly boundary conditions, significantly enhances predictive reliability, reduces worst-case errors, and ensures the model respects the physical bounds of the system. This makes the "Data, PDE and BCs loss" configuration the most suitable for practical applications where accurate range coverage and error minimisation are critical.

Figure 8 provides a visual comparison of two machine learning approaches for modelling LST: a PINN with only data loss and a RF with 1000 estimators. Both are evaluated against ground truth data (27.7°C to 44.2°C), which shows characteristic spatial patterns of warmer regions (red) and cooler linear features like roads or rivers (yellow).

The PINN-modelled LST ($27.9^\circ\text{C}\text{--}45.5^\circ\text{C}$) replicates the spatial structure well but exhibits a positive bias, overestimating the maximum temperatures. Conversely, the RF-predicted LST ($28.5^\circ\text{C}\text{--}42.5^\circ\text{C}$) underestimates peak temperatures, showing a negative bias, though it maintains the core spatial patterns. Analysis of absolute errors reveals that the PINN's errors are more localised and severe in some areas (0.00°C to 7.65°C), while the RF's error range is narrower (0.00°C to 6.69°C), indicating slightly better handling of extreme cases. The histograms further illustrate this: the PINN's temperature distribution extends into warmer values than observed, while the RF's distribution is too narrow and fails to capture the upper tail of temperatures.

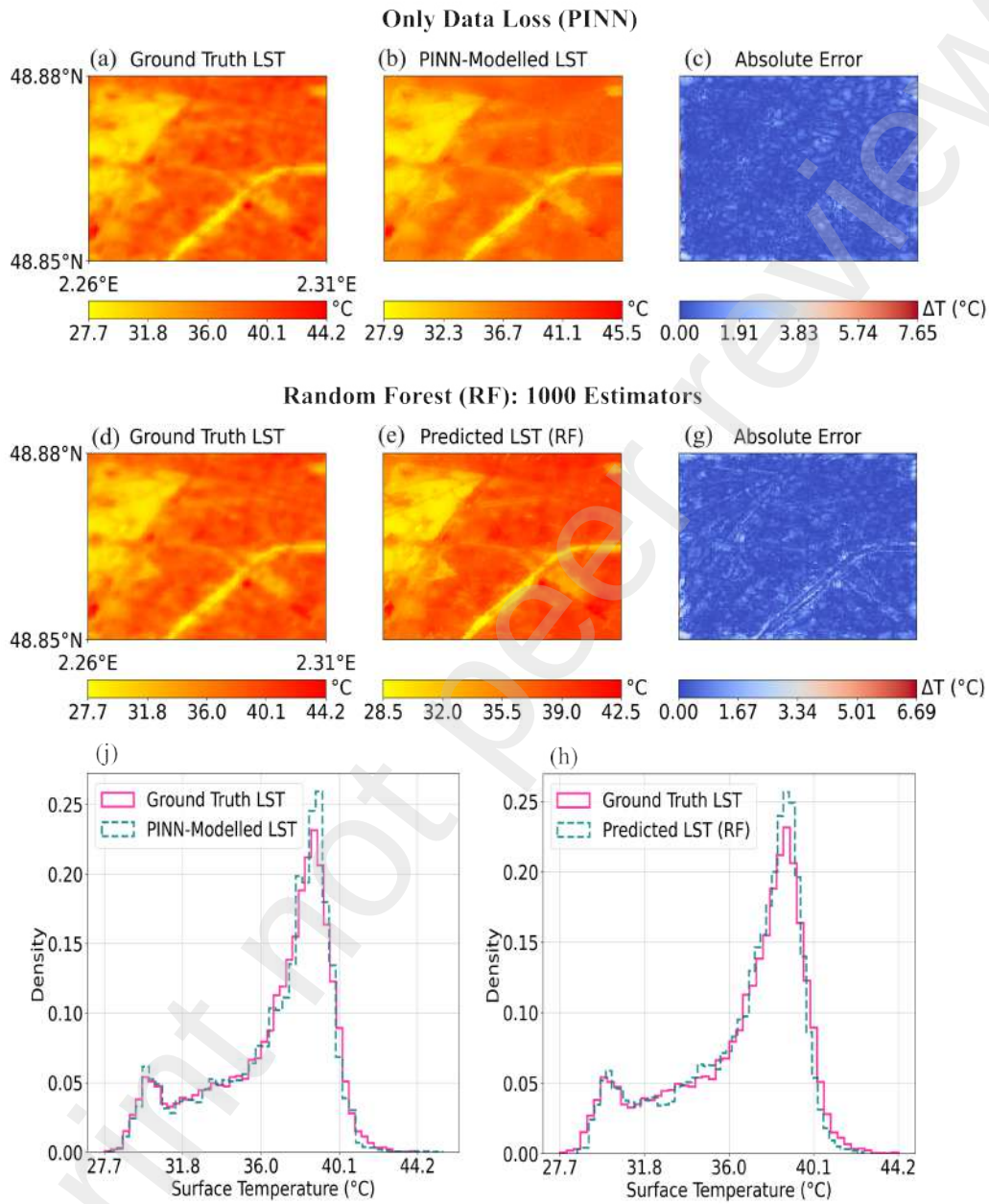


Figure 8: Comprehensive comparison of LST predictions using a PINN (with only data loss) and a Random Forest (RF) with 1000 estimators. Panels (a)–(c) and (d)–(g) display spatial LST maps, while (h) and (j) show the corresponding temperature distribution histograms.

Ultimately, the PINN with only data loss introduces a warm bias, whereas the RF, while better at minimising the worst deviations, consistently underestimates high temperatures. This contrast underscores the fundamental trade-offs in model architecture between spatial fidelity and error distribution for LST prediction.

4.3. Application to Green Infrastructure Planning

Building on its predictive accuracy, we demonstrate the model’s utility as a practical decision-support tool for urban planners by simulating the cooling effects of green roof interventions [60, 61]. This application quantifies the potential temperature reductions for buildings located more than 15 pixels away from railways.

4.3.1. Identifying Thermal Hotspots and Designing Intervention Scenarios

To target interventions effectively, we first identified UHI hotspots using observed LST data [62] and the Urban Heat Thermal Index (UHTI), as defined in Equation 13. A percentile-based classification system was employed for this analysis.

$$\text{UHTI} = \frac{\text{LST} \times (1 - \text{NDVI})}{h_{\text{safe}}} \quad (13)$$

where $h_{\text{safe}} = \max(h, 0.01)$ is a safe emissivity value, and h represents the land surface emissivity to ensure numerical stability.

Table 4: Thermal hotspot and coolspot classification scheme

Thermal Class	Level	Percentile	LST Threshold(°C)	UHTI Threshold (°C)
Hotspot	1	95th	≥ 40.15	≥ 37.92
	2	90th	≥ 39.69	≥ 36.47
	3	85th	≥ 39.36	≥ 35.17
Coolspot	1	15th	≤ 32.59	≤ 4.65
	2	10th	≤ 31.28	≤ 8.75
	3	5th	≤ 30.08	≤ 11.77

As detailed in Table 4, the classification framework categorises areas into three intensity levels for both thermal hotspots and coolspots, based on percentile thresholds for LST and UHTI. The most severe hotspots (Level 1, defined by the 95th percentile and an LST $\geq 40.15^\circ\text{C}$) represent the core of

the UHI. In contrast, the most pronounced coolspots (Level 1) are identified by the 15th percentile.

Figure 9 presents a comparative spatial analysis of thermal hotspots and green roof prioritisation strategies within a study area of Paris. Panel (a) maps LST intensity, where cool areas (light to dark blue) and hot areas (light pink to dark red) represent percentile-based thermal extremes (e.g., Hot95 corresponds to the top 5% hottest areas). The results show distinct spatial patterns: cool zones align prominently with the Seine River and large green spaces, while hotspots cluster in dense urban fabric, particularly in the eastern and central districts. The river's role as a natural cooling corridor is clearly visible, contrasting with the elevated temperatures of built-up areas.

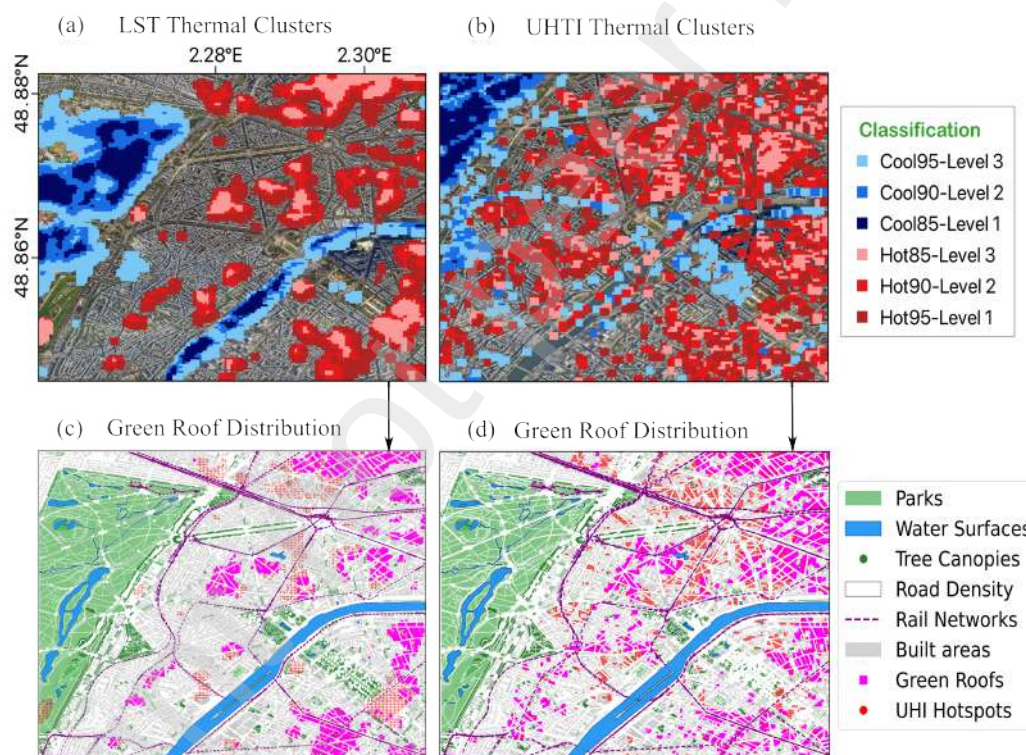


Figure 9: Targeting green roof interventions using LST-based and UHTI-based thermal hotspot analysis in Paris. (a) LST hotspot intensity levels; (b) UHTI hotspot intensity levels; (c–d) Priority buildings for green roof placement derived from LST-based and UHTI-based analysis, respectively.

Panel (b) displays the UHTI, a metric that incorporates urban-specific

factors such as building density and materials. While the colour scheme is consistent with Panel (a), the UHTI reveals a more concentrated and widespread distribution of hotspots. This suggests it captures nuanced thermal vulnerabilities inherent to urban form, with cool zones appearing less extensive than in the LST map.

Panels (c) and (d) illustrate green roof prioritisation derived from the LST and UHTI analyses, respectively. The priorities in Panel (c) align with the LST hotspots from Panel (a), targeting dense urban zones with the highest surface temperatures. On the other hand, the priorities in Panel (d) shift to match the UHTI hotspots from Panel (b), highlighting different areas, such as those near rail networks. This divergence demonstrates how the choice of thermal metric directly shapes intervention strategies: LST targets peak temperatures, while UHTI identifies areas where urban structure amplifies heat risk.

In combination, Figure 9 demonstrates that integrating multiple thermal metrics, using LST for direct temperature and UHTI for urban heat dynamics, enables a more nuanced approach to optimising green infrastructure. This comparative analysis provides a robust framework for refining climate adaptation strategies in Paris.

4.3.2. Biophysical Profiling of Targeted Buildings

Figure 10 provides a statistical characterisation of the biophysical properties of buildings targeted for green roofs in Figure 9, assessing their suitability through paired histograms and kernel density estimates (KDEs) for NDVI and emissivity.

For LST-based targets (a–b), the NDVI distribution (a) shows a median of 0.13, with a concentrated KDE peak between 0.1–0.2. This indicates a focus on areas with moderate existing vegetation, suggesting LST prioritises sites where green roofs could enhance pre-existing cooling. Conversely, UHTI-based targets (c) have a lower median NDVI of 0.08 and a broader distribution skewed towards near-zero values, peaking around 0.05. This reflects a focus on barren, highly urbanised zones where new vegetation would provide the greatest marginal cooling benefit.

Both methods, however, select buildings with similarly high emissivity (median ~ 0.97 in b and d), confirming that thermally efficient surfaces common in dense urban fabric are a shared prerequisite.

This divergence in NDVI profiles underscores a critical methodological distinction. LST-based targeting leverages incremental greening in moderately

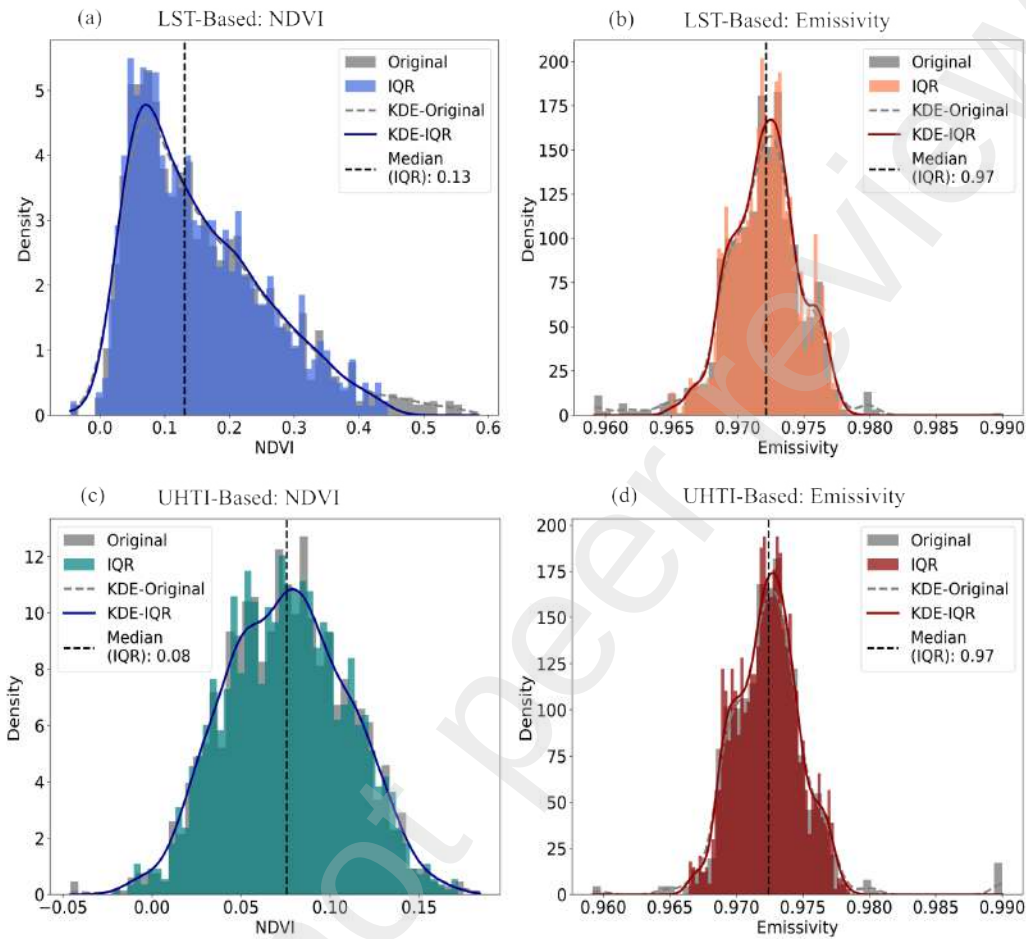


Figure 10: NDVI and emissivity distributions for buildings targeted using LST-based (a-b) and UHTI-based (c-d) selection methods.

vegetated hotspots, while UHTI prioritises transformative intervention in structurally vulnerable, vegetation-scarce areas. The analysis confirms that the selected buildings are systematically aligned with each metric's logic, with UHTI's focus on low-NDVI sites indicating a higher potential for direct heat island mitigation. Meanwhile, the LST-based approach offers a practical pathway for implementation by building upon existing vegetative cover.

4.3.3. Quantifying the Cooling Effects of Green Roofs

To simulate the cooling effects of green roofs, we modified input parameters for targeted buildings based on empirical measurements from extensive green roofs in temperate European cities. These typically exhibit NDVI values of 0.6-0.8 during peak growing season [63], so we set NDVI to 0.75 with corresponding emissivity adjustments. The underlying urban morphology, represented by the 'Building' binary layer, remained unchanged. This conservative approach means the model continues to partially interpret these pixels as built surfaces, likely resulting in a slight underestimation of the full cooling potential achievable with more comprehensive surface modifications. We then applied the trained PINN to this modified dataset, calculating temperature reduction as $\Delta T = T_{\text{green roof}} - T_{\text{original prediction}}$.

Figure 11 quantifies the cooling impact of deploying green roofs on the buildings identified through the LST-based thermal hotspot analysis (Figure 9a,c), providing a comprehensive spatial and statistical evaluation.

Panels (a) and (b) show the baseline PINN-predicted LST and the simulated LST after green roof implementation, respectively, using an identical temperature scale (27.7-43.9°C) to emphasise spatial changes. The baseline map (a) reveals intense thermal hotspots concentrated in dense urban areas, while the post-intervention map (b) shows a marked reduction in these high-temperature zones, particularly where buildings were targeted. This visual shift is quantified in panel (c), which maps the spatial cooling effect (ΔT). Green hues indicate temperature reductions of up to -4.86°C, predominantly localised over the targeted building footprints.

The statistical distributions in panels (d) and (e) further validate the cooling effect. The baseline LST histogram (d) peaks at 35.8°C, encompassing 62.4% of pixels, with 5.8% exceeding 39.9°C. Following the intervention, the histogram (e) shows a higher peak at the same temperature (64.2% of pixels) and reduces the high-temperature tail, with only 3.9% of pixels remaining above 39.9°C. This demonstrates a systemic mitigation of extreme heat.

Critically, panel (f) quantifies the frequency distribution of ΔT , revealing that 96.2% of pixels experience cooling. The majority of this cooling (82.6%) is concentrated in the -1.22°C bin. This near-universal effect, coupled with a 1.9 percentage point reduction in pixels exceeding 39.9°C (evident in panels d-e), underscores the efficacy of green roofs in alleviating urban heat stress.

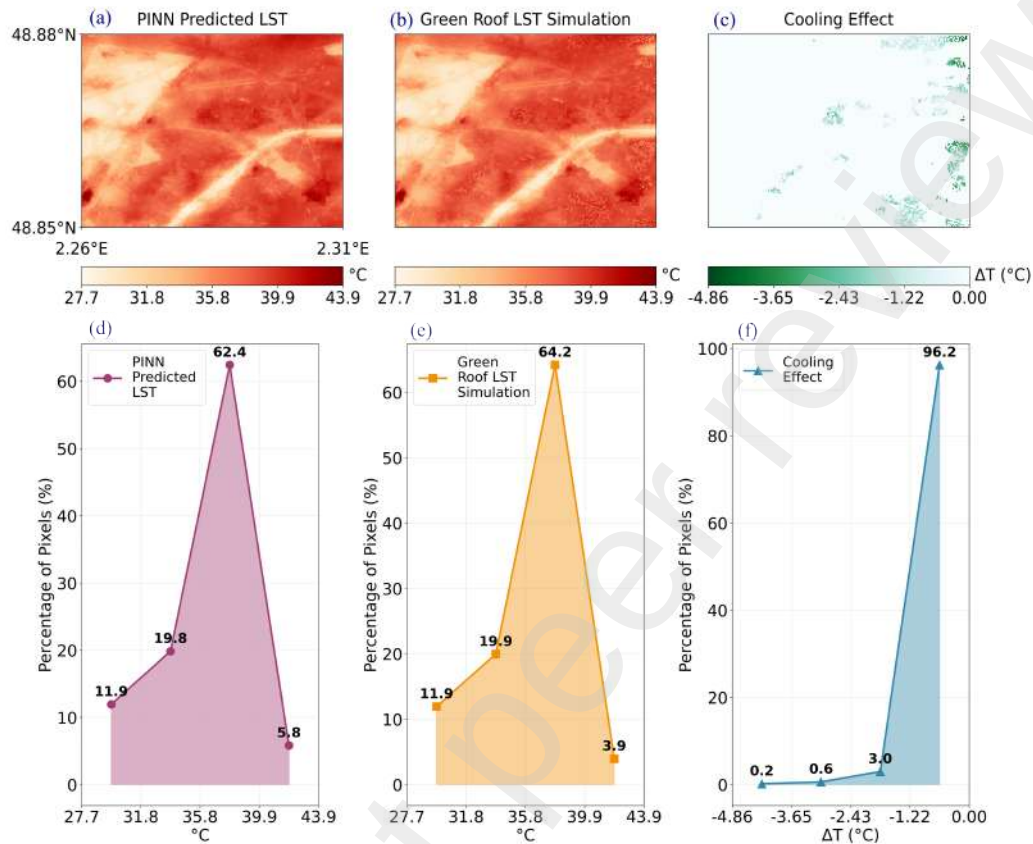


Figure 11: Assessment of green roof cooling effects for LST-based targets. (a) Baseline LST; (b) Modified LST; (c) Cooling magnitude (ΔT); (d,e) Pre- and post-intervention distributions; (f) Cooling effect frequency.

This analysis confirms that targeted interventions based on LST-derived hotspots yield measurable and spatially coherent temperature reductions. The statistical shift towards lower thermal extremes provides empirical support for green infrastructure deployment. By integrating PINN-based prediction with spatial and distributional analysis, the results demonstrate that green roofs not only cool localised areas but also reshape the city's thermal profile, reducing the frequency and intensity of extreme heat events in vulnerable urban zones.

Figure 12 evaluates the cooling effects of green roofs deployed on buildings identified through UHTI-based thermal hotspots (Figure 9b,d). This parallel assessment enables a direct comparison of how different targeting strategies influence cooling outcomes.

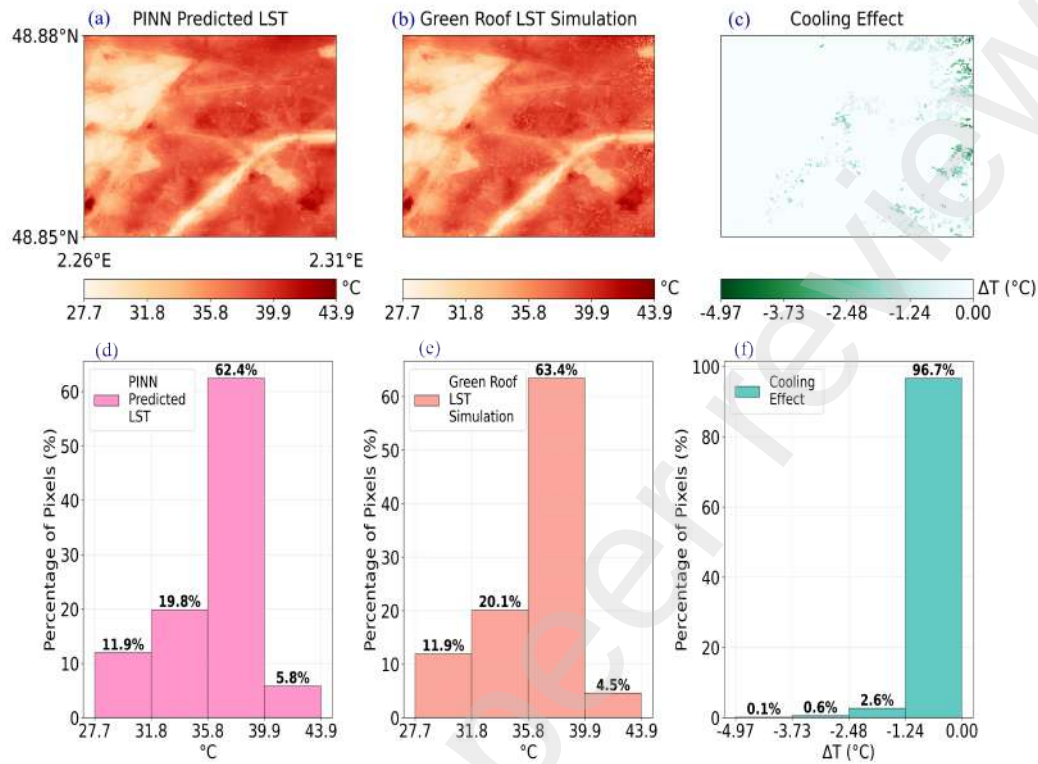


Figure 12: Evaluating UHTI-based intervention efficacy. (a) Pre-implementation LST; (b) Simulated conditions with green roofs; (c) Cooling magnitude mapping; (d,e) Statistical comparison of LST; and (f) Cooling impact assessment.

The analysis follows the same structure as Figure 11, presenting baseline and post-intervention LST maps alongside spatial cooling patterns and statistical distributions. The baseline conditions show comparable thermal stress, with 5.8% of pixels exceeding 39.9°C, identical to the LST-based baseline in Figure 11a. Following green roof implementation, the spatial cooling pattern in Figure 12c reveals a maximum temperature reduction of -4.97°C, slightly greater than the -4.86°C observed in Figure 11c. However, the cooled areas appear more fragmented, reflecting UHTI's focus on structurally vulnerable, lower-NDVI zones as established in Figure 10c.

Statistically, the intervention produces a similar shift in the temperature distribution, though with nuanced differences. The reduction in extreme heat (Figure 12e) is somewhat smaller than in Figure 11e, with 4.5% of pixels remaining above 39.9°C compared to 3.9% in the LST-based approach. This suggests UHTI-targeted areas experience marginally less mitigation of the

hottest zones. Conversely, the cooling effect distribution in panel (f) shows that 96.7% of pixels experience cooling, with the majority concentrated in the -1.24°C bin, indicating a slightly more uniform cooling effect than achieved through LST-based targeting.

This comparative analysis reveals that the choice of thermal metric creates distinct cooling outcomes. UHTI-based interventions, targeting areas with minimal existing vegetation, introduce new cooling capacity across a broader area. Whereas, LST-based approaches, focusing on zones with moderate vegetation, achieve sharper reductions in extreme temperatures by amplifying existing cooling mechanisms. Together, these findings demonstrate the complementary value of both targeting strategies: while LST prioritisation more effectively suppresses thermal extremes, UHTI targeting delivers more consistent cooling across vulnerable urban structures. This duality highlights the importance of metric selection based on specific intervention goals, whether addressing peak temperatures or broader heat vulnerability.

4.4. Study Limitation

Despite its strong predictive performance, the framework presented here has several limitations that warrant consideration for both interpretation and future development:

First, spatial and temporal resolution constraints: The model's spatial resolution is constrained to 30 metres by its satellite-derived inputs. While suitable for neighbourhood-level analysis, this resolution inevitably masks critical microclimatic variations within urban canyons, adjacent to building walls, and in small, shaded courtyards. This coarse granularity can introduce systematic bias by averaging temperatures across heterogeneous surfaces; for example, it may fail to capture the full cooling effect of a small shaded area while underestimating the peak temperature of a sunlit, impervious surface within the same pixel. Furthermore, while this study and a prior application [34] demonstrate the model's strong spatial generalisability across different cities, both validations use single temporal snapshots taken from summer conditions. The framework's performance across different seasons or under varying synoptic conditions therefore remains an area for future investigation.

Second, physical formulation simplifications: The model relies on a simplified 2D, steady-state heat equation to ensure computational stability and integration within the PINN framework. While effective for capturing dominant diffusion processes at neighbourhood scales, this approach cannot represent more complex, dynamic phenomena. It excludes critical processes such as

3D radiative trapping, wind-driven advection, and the complete diurnal cycle of heat storage and release, all of which are pivotal for understanding the full temporal dynamics of urban heat islands [51, 64]. The omission of these processes, particularly 3D radiative effects, means the model is likely to systematically underestimate the absolute intensity of the UHI, especially within dense urban cores where deep street canyons trap heat. The parameterization of complex phenomena like urban canyon effects using simplified proxies (e.g., building perimeter-to-area ratio) further constrains the model’s physical completeness.

Third, context specificity and transferability: The model’s parametrisation and validation are inherently specific to the urban fabric and climate of Paris. We cannot assume its transferability to cities with radically different morphologies, such as low-rise, sprawling metropolises, or to contrasting climatic regimes without substantial recalibration. Several components are particularly location-specific: the empirically informed coefficients in the heat source Q_{Sources} or sinks Q_{Sinks} terms are calibrated to the material properties and typical urban morphology of Paris, and the chosen boundary conditions reflect its unique geographical context. Moreover, the relationships learned between input features and land surface temperature may not hold under different background climates or prevailing wind patterns. Applying this framework elsewhere would therefore require not only equivalent local geospatial data but also a thorough re-evaluation of its core assumptions and parametrisations.

The study identifies a key gap in the current model’s capabilities, particularly concerning resolution, temporal cycles, and generalisability. Addressing this gap, as outlined in the Future Outlook (Section 5), involves plans to incorporate higher-resolution 3D urban data, extend the model to simulate diurnal and seasonal cycles, and validate it across diverse urban and climatic contexts.

5. Conclusions and Future Outlook

This work has developed and validated a novel PINN framework that advances urban heat island modelling by seamlessly integrating data-driven learning with fundamental thermodynamic principles. The core methodological contribution demonstrates how a reproducible workflow for fusing heterogeneous geospatial data into a cohesive input structure can be effectively integrated within a PINN architecture, addressing a significant bottleneck

in urban climate analytics. Applied to a case study of Paris during summer conditions, the framework demonstrated exceptional predictive accuracy ($R^2 = 0.9741$) while ensuring physical consistency with a simplified urban energy balance model, establishing a reliable and scalable approach for thermal environment analysis at neighbourhood scales.

Beyond accurate spatial prediction, this research bridges a critical gap between high-fidelity modelling and practical urban planning applications. By simulating targeted green roof scenarios, we transition from passive observation to active intervention planning. The framework offers an actionable methodology for prioritising limited resources, demonstrating that strategically deployed nature-based solutions can yield localised cooling of up to 4.86°C with LST-based targeting or 4.97°C with UHTI-based targeting in critical hotspots. This capacity to quantitatively forecast intervention efficacy before implementation represents a significant stride toward more resilient and sustainable smart city development.

The limitations identified here define clear priorities for future research, which will extend the framework in the following directions:

- Temporal dynamics and generalisation: A critical next step is to move beyond single-snapshot validation by incorporating diurnal and seasonal cycles. This will involve leveraging time-series satellite data and introducing a time-dependent term into the governing equation, directly addressing the current limitation by validating the model's ability to capture the evolution of urban temperatures across different temporal scales [29].
- Enhancing physical realism through 3D geometry: Explicitly integrating 3D urban geometry to model radiative trapping within street canyons and advective processes. Using building height data from LiDAR and detailed urban morphology parameters would better capture street-level microclimates and the complex interplay between urban form and temperature distribution [65].
- Ensuring transferability and robust validation: Applying and validating this framework in cities with contrasting climates (e.g., arid, tropical) and urban forms (e.g., low-rise sprawl) [66]. Establishing robust validation protocols across diverse city typologies will be essential for enhancing the model's global applicability and identifying universal parameters alongside context-specific adaptations [67].

- Expanded intervention strategies: Further research will explore additional mitigation strategies beyond green roofs, such as cool materials, urban forestry, and water-based cooling systems [68]. The framework could also be extended to incorporate socio-economic factors and energy consumption patterns [69], enabling more holistic urban planning assessments that consider both environmental and human dimensions of UHI.

Ultimately, the present study establishes a foundational methodology that effectively balances physical principles with data-driven learning for urban climate analysis. While current limitations define clear boundaries for application, the strategic directions outlined provide a roadmap for transforming this promising framework into a comprehensive, temporally-aware, and practically accessible tool for evidence-based urban heat mitigation and climate adaptation planning worldwide.

Appendix A. Weight Configuration Scenario Results

The spatial patterns in Figure A1 (Scenarios 1–5) align closely with the performance metrics in Table A1, demonstrating how the assigned weights for heat sources and sinks shape thermal dynamics.

Scenario 1 (enhanced cooling) employs balanced source/sink weights (e.g., $BD = 0.25$, $PD = 0.20$), producing moderate heat source intensity (max = 0.60 in "combined heat sources" of Figure A1) alongside the highest sink contribution (max = 0.43 in "combined heat sinks"). This balance manifests in the balanced Q panel's mixed red/blue hues (range: -0.30 to 0.56) and corresponds to the 78.1/21.9% source/sink ratio in Table A1. The configuration yields a strong model fit ($R^2 = 0.9683$, $RMSE = 0.5552$), with predicted temperatures (27.9–44.4°C) closely matching observations (27.7–44.2°C).

By comparison, scenario 2 (extreme urban heat) applies elevated source weights ($BD = 0.40$, $RD = 0.35$) with minimal sink weights ($CP = 0.04$), generating intense source signals (max = 1.03) against weak sinks (max = 0.21). The result is a fully red balanced Q panel (0.01–1.00) showing complete source dominance (100.0/0.0%). Despite high R^2 (0.9666), the wider predicted range (28.1–45.0°C) and elevated max error (5.50°C) indicate overestimation in extreme heat zones.

Shifting focus to transportation, scenario 3 assigns high weight to road density ($RD = 0.40$), exhibiting moderate source intensity (max = 0.81) while

achieving the lowest RMSE (0.5282) and MAE (0.3876), indicating superior accuracy. Despite 99.9/0.1% source dominance (balanced Q range: -0.11 to 0.75), faint blue patches reveal minor sink influence.

Similarly building-focused, scenario 4 emphasizes building density (BD = 0.50), displaying the strongest source intensity in built-up areas (max = 0.75). With a balanced Q ratio of 79.5/20.5% and RMSE (0.5558) comparable to scenario 1, this configuration maintains relative balance despite the building emphasis.

Finally, scenario 5 (water-focused cooling) prioritizes water density (WD = 0.30), enhancing sink intensity (max = 0.39) and creating localized blue regions in the balanced Q panel. Nevertheless, source dominance persists (96.3/3.7%), and this scenario shows the highest max error (6.76°C), likely due to uneven spatial cooling effects.

Cumulatively, the colour gradients and spatial heterogeneity in these figures directly illustrate the consequences of the weight distributions in Table A1. They show how targeted interventions such as increasing water features or managing building density, can modulate urban heat and influence the reliability of the model's predictions.

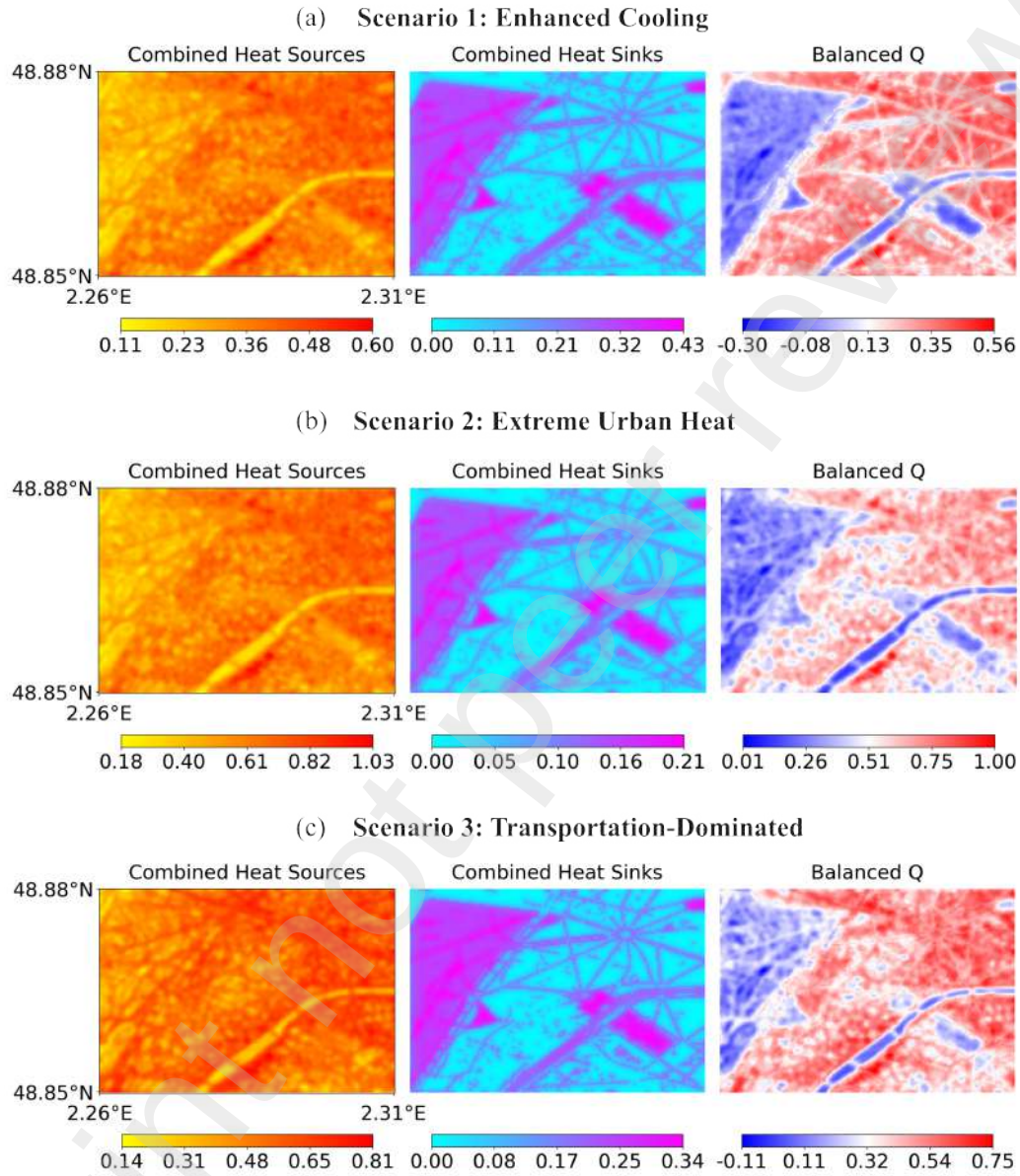


Figure A1: Spatial distribution of thermal metrics for scenarios 1–3: (a) enhanced cooling, (b) extreme urban heat, and (c) transportation-dominated. Each scenario's row shows the 'combined heat sources' (yellow-red), 'combined heat sinks' (blue-magenta), and 'balanced Q' (blue-red) maps. The visualised patterns demonstrate the transition from a balanced thermal landscape in (a) to extreme source dominance in (b), and the accurate, transport-driven heating in (c), as quantified in Table A1.

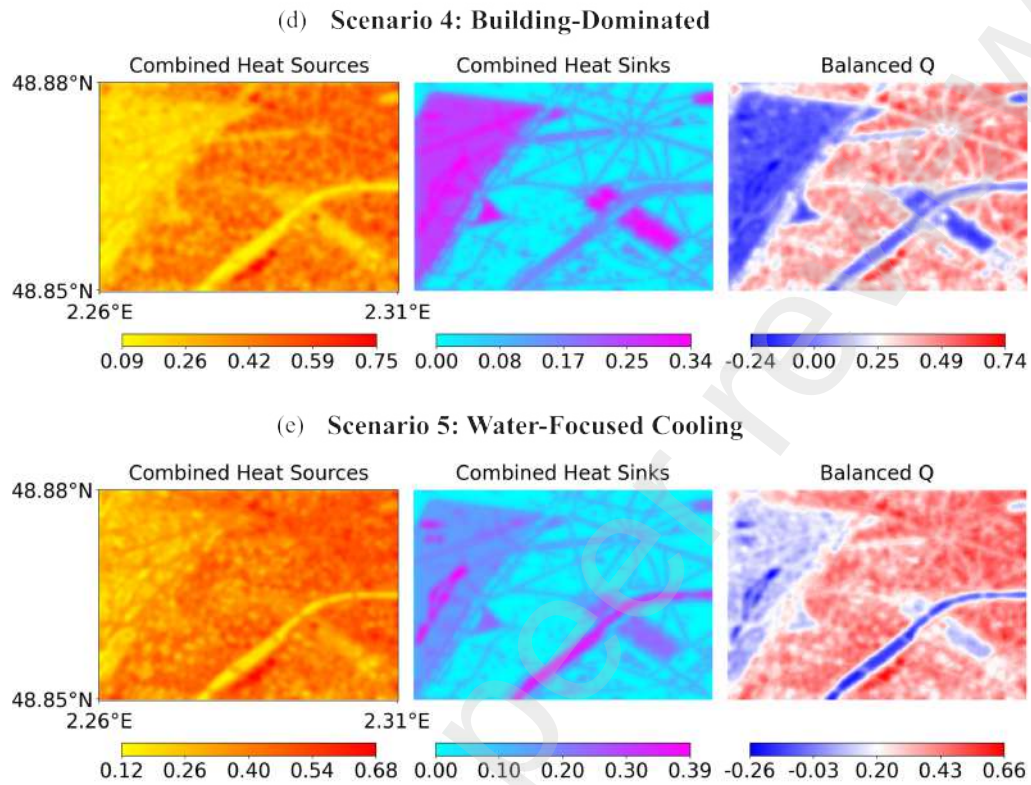


Figure A1 (continued). Spatial distribution of thermal metrics for scenarios 4–5: (d) building-dominated and (e) water-focused cooling.

Table A1: Scenario Analysis Results

Scenario	Sources Weights	Sinks Weights	Balanced Q (Sources/Sinks)	R ²	RMSE (°C)	MAE (°C)	Predicted Range (°C)	Max Error (°C)
1: Enhanced Cooling	BD: 0.25, RD: 0.20, IDR: 0.15, UC: 0.10	PD: 0.20, WD: 0.15, TD: 0.15, CP: 0.10	78.1/21.9%	0.9683	0.5552	0.4099	27.9–44.4	5.92
2: Extreme Urban Heat	BD: 0.40, RD: 0.35, IDR: 0.25, UC: 0.20	PD: 0.10, WD: 0.08, TD: 0.08, CP: 0.04	100.0/0.0%	0.9666	0.5696	0.4211	28.1–45.0	5.50
3: Transport-Dominated	BD: 0.20, RD: 0.40, IDR: 0.25, UC: 0.10	PD: 0.15, WD: 0.10, TD: 0.10, CP: 0.10	99.9/0.1%	0.9713	0.5282	0.3876	27.5–44.8	4.16
4: Building-Dominated	BD: 0.50, RD: 0.15, IDR: 0.10, UC: 0.10	PD: 0.20, WD: 0.10, TD: 0.10, CP: 0.05	79.5/20.5%	0.9682	0.5558	0.4079	27.5–44.9	5.83
5: Water-Focused Cooling	BD: 0.30, RD: 0.25, IDR: 0.15, UC: 0.10	PD: 0.10, WD: 0.30, TD: 0.10, CP: 0.05	96.3/3.7%	0.9694	0.5453	0.4056	27.5–44.8	6.76
6: Baseline (Current Implementation)	BD: 0.30, RD: 0.25, IDR: 0.20, UC: 0.15	PD: 0.15, WD: 0.10, TD: 0.10, CP: 0.10	97.6/2.4%	0.9741	0.5019	0.3723	27.7–43.9	4.53
True Range (all scenarios): 27.7–44.2°C								

Abbreviations: BD: Building Density, RD: Road Density, IDR: Inverse Distance Roads, UC: Urban Canyon, PD: Park Density, WD: Water Density, TD: Tree Density, CP: Cooling Proximity

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used Grammarly in order to improve language, clarity, and readability. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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